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About the UltraWave 12G MWModulatedSource : Features

Features

Basic functional specifications for UltraWave 12G MWModulatedSource are:

- Frequency range: 76MHz to 6GHz

- Modulation bandwidth: 80MHz
- Amplitude range:
  - 76MHz to 2.5GHz: -130dBm to +13dBm
  - 2.5GHz to 6GHz: -130dBm to +10dBm
- Includes I and Q AWGs to support analog and digital modulation schemes
- Supports IQ signal definition in TSD files
- Supports IF signal definition for Gen4 MW compatibility
- Amplitude control of the modulated RF signal
- Signal power adjustment based on the crest factor of the modulation signal

For UltraWave 12G hardware specifications, see UltraWave 12G Specifications.

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About the UltraWave 12G MWModulatedSource : Safety Information

Safety Information

When operating the Microwave Modulated Source instrument, always follow general UltraFLEX test system safety guidelines. See Test System Safety.

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About the UltraWave 12G MWModulatedSource : Key Terms

Key Terms

AWG	Arbitrary Waveform Generator
baseband	Low frequency information contained in a modulated signal
I signal	In-phase (real) component of a baseband signal
IF	Intermediate Frequency
LO	Local Oscillator
Q signal	Quadrature (imaginary) component of a baseband signal
RF	Radio Frequency

For more MW terms and definitions, see Key Terms in About the UltraWave 12G Subsystem.

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About the UltraWave 12G MWModulatedSource

About the UltraWave 12G MWModulatedSource

The MWModulatedSource instrument generates modulated signals in the RF frequency band in the frequency range of 76MHz to 6GHz. Applications for the Microwave Source include prescalers, RF amplifiers, TV tuners, cellular radio, IF devices, and operational amplifiers.

This section includes the following topics:

- Features
- Safety Information
- Key Terms

Microwave Modulated Source information includes the following sections:

- UltraWave 12G MWModulatedSource Theory of Operation
- UltraWave 12G MWModulatedSource Programming
- UltraWave 12G MWModulatedSource Debug Display

Related Topics

- UltraWave 12G Specifications

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UltraWave 12G MWModulatedSource Debug Display

UltraWave 12G MWModulatedSource Debug Display

The Microwave Debug Display can display one or more instrument blocks for the MWModulatedSource logical instrument. Each instrument block shows the state of an instrument in the context of a site.

This section includes the following topics:

- MW12G MWModulatedSource Debug Display Controls

For an overview of all displays supporting the MW subsystem, see UltraWave 12G Displays.

For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays

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UltraWave 12G MWModulatedSource Debug Display : MW12G MWModulatedSource Debug Display Controls

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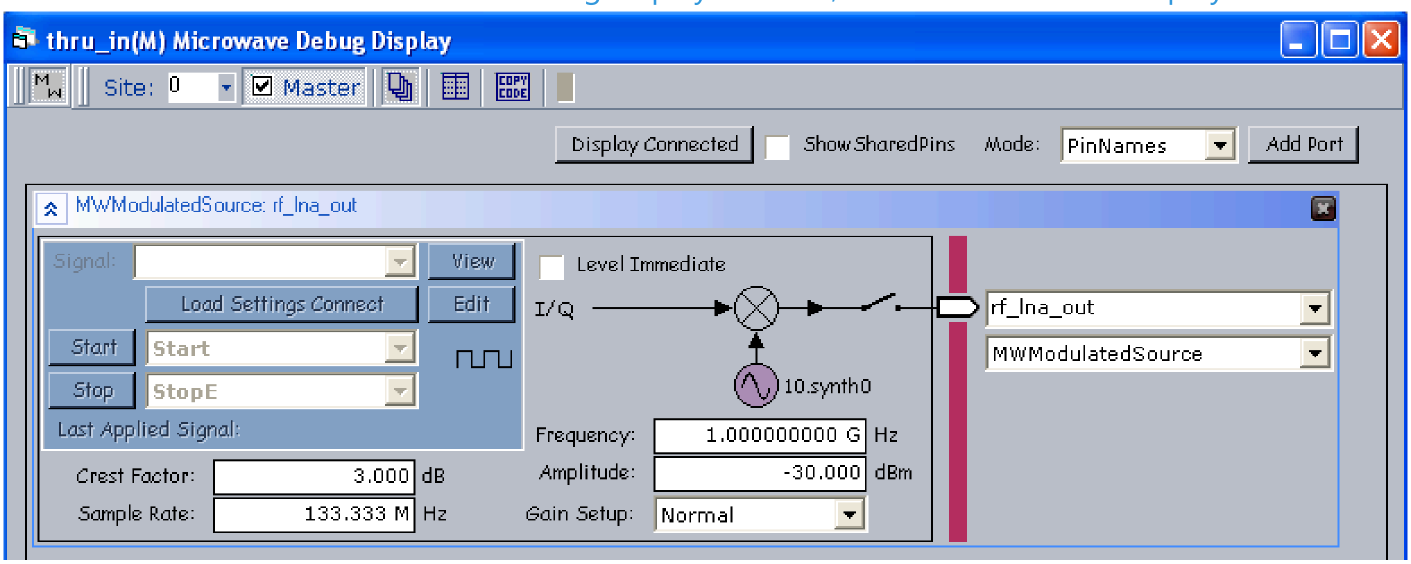
MW12G MWModulatedSource Debug Display Controls

MWModulatedSource port display controls allow you to view and set instrument parameters, which are also accessible through the Pins interface of the MWModulatedSource VBT language.

For information about the common debug display controls, see UltraWave 12G Displays.



MW12G MWModulatedSource Logical Instrument Display

The following controls are available on the MWModulatedSource logical instrument block:

Frequency	Displays and sets the modulated signal frequency. The tooltip displays the allowable frequency range.
Amplitude	Displays and sets the modulated signal amplitude. The tooltip displays the allowable amplitude range.

Level Immediate	Sets the MWModulatedSource leveling mode to Level Immediate, when checked. See MWModulatedSource.Pins.LevelImmediate in the VBT Syntax Reference.
Sample Rate	Displays and sets the modulator sample rate for the modulation signal. The tooltip displays the allowable sample rate range.
Crest Factor	Displays and sets the crest factor for a modulated signal. The tooltip displays the allowable crest factor range.
Gain Setup	Displays and sets the gain setup for a modulated tone. The drop-down menu includes: • Normal • LowDistortion
synth label	Identifies the synthesizer (slot and number) used by this MWModulatedSource logical instrument.
Signal	Displays the selected source Signal on a drop-down list of available signals.
Load Settings Connect	Loads the selected signal and connects the MWModulatedSource.
View	Displays a list of signal viewer tools (such as WaveScope or VectorScope) for the selected Signal on a drop-down list of available viewers.
Info	Displays property values for the selected Signal in a pop-up balloon window.
Start	Starts the selected Signal using one of the start options: • Start • Start1 • StartE • Start1E

Stop	Stops the selected Signal using one of the stop options: • Stop • StopE
Last Applied Signal	Displays the name of the last Signal applied to the MWModulatedSource.
connection relay	The relay control connects or disconnects the MWModulatedSource logical instrument to the specified pin or channel. Click the relay to open or close the connection.

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UltraWave 12G MWModulatedSource Programming

UltraWave 12G MWModulatedSource Programming

The MWModulatedSource is a MW logical instrument in the IG-XL environment. You add the MWModulatedSource to an IG-XL program in the same general way that you add other MW instruments.

The following topics are available:

- Sourcing Modulated Signals Using the MWModulatedSource
- MWModulatedSource Pattern Control

Before writing test code using the MWModulatedSource, see UltraWave 12G Instrument Programming.

For programming concepts, a procedure, and an example, see Sourcing Modulated Signals Using the MWModulatedSource.

For more information, see:

- MWModulatedSource (VBT Syntax Reference)
- UltraWave 12G Examples

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UltraWave 12G MWModulatedSource Programming : Sourcing Modulated Signals Using the MWModulatedSource

Sourcing Modulated Signals Using the MWModulatedSource
This section includes the following topics:

- Programming Concepts
- Programming Steps

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UltraWave 12G MWModulatedSource Programming : Sourcing Modulated Signals Using the MWModulatedSource : Programming Concepts

Programming Concepts

TSD File Support

The MWModulatedSource accepts signal definitions for modulated signals in TSD files. TSD files are loaded using the Pins.Signals.SignalDefinitionFile property.

For more information, see Using the ESA Toolkit.

MWModulatedSource Reset Values

The table lists the local reset values for the MWModulatedSource.

MWModulatedSource Reset Values

Setting	Reset Value
Amplitude	-30dBm
Frequency	1.0GHz
Sample Rate	133.333MHz
Crest Factor in dB	3.00
GainSetup	Normal
Connections	Disconnected

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UltraWave 12G MWModulatedSource Programming : Sourcing Modulated Signals Using the MWModulatedSource : Programming Steps

Programming Steps
To source a modulated signal to an assigned RF port:

1.	Create a modulation signal in VBT or import a signal waveform from an external file.

Use the Signals interface for the AWG instrument to specify all parameters of your modulation signal. Use the Wave Definitions sheet to specify your wave shape.

2.	Specify the crest factor. Set up the MWModulatedSource with:

–	Amplitude
–	Frequency
–	IF Frequency
–	Crest factor

3.	Connect the MWModulatedSource.

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UltraWave 12G MWModulatedSource Programming : MWModulatedSource Pattern Control

MWModulatedSource Pattern Control

You can control an MWModulatedSource instrument using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and Signals programming.

You create a pattern independently of the test program using the PatternTool or a text editor. Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well the pattern microcode used for controlling the instrument during pattern execution.

The following topics are available:

- MWModulatedSource Pattern Microcodes
- MWModulatedSource ASCII Pattern Syntax

☑ **Note:**

To use pattern microcodes to control an MWModulatedSource on UltraFLEX, you must include a digital pin in the pin map, the channel map, and the pin list of the pattern module.

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UltraWave 12G MWModulatedSource Programming : MWModulatedSource Pattern Control : MWModulatedSource Pattern Microcodes

MWModulatedSource Pattern Microcodes

A microcode controls a single function, such a starting or stopping the sourcing of a signal. Extended microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a signal is sourced. To establish repeatable timing, or to synchronize with other instruments (such as the digital subsystem) in the tester, use a pattern. If you want to start generating a signal but are not concerned with precise timing, however, then you do not need a pattern.

The table lists the extended microcodes used to issue commands to an MWModulatedSource instrument from a pattern.

MWModulatedSource Pattern Microcodes

Microcode	Functional Description
Nop	No operation.
Resync	Forces phase alignment of an MWModulatedSource instrument clock with the HSD T0 clock. See Use of Resync.

Start <SignalName>	The instrument immediately (on the next available logical analog clock) starts sourcing the named signal in SMEM and repeats it continuously until a Stop microcode is executed, the instrument is reset, or another signal is started. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used (see TheHdw.MWModulatedSource.Pins.Signals.DefaultSignal).
StartE <SignalName>	At the end of the currently sourcing signal's segment, the instrument sources the named signal in SMEM and repeats it continuously until a STOP microcode is executed, the instrument is reset, or another signal is started. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used (see TheHdw.MWModulatedSource.Pins.Signals.DefaultSignal).
Start1 <SignalName>	The instrument immediately starts sourcing (on the next available logical analog clock) the named signal in SMEM and sources it exactly once. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used (see TheHdw.MWModulatedSource.Pins.Signals.DefaultSignal).
Start1E <SignalName>	At the end of the currently sourcing signal's segment, the instrument sources the named signal in SMEM and sources it exactly once. This microcode takes an optional argument, which is the name of the signal to be sourced.If the signal name is not specified, the default signal is used (see TheHdw.MWModulatedSource.Pins.Signals.DefaultSignal).
Stop	The instrument stops sourcing the currently sourced signal immediately. STOP does not impact a signal that is running once.
StopE	The instrument stops sourcing the currently sourced signal at the end of the signal.
LoadSettingsConnect <SignalName>	Instructs an MWModulatedSource to load the settings for the named capture signal and connects the hardware to the DUT.

	SignalName is required.
Disconnect	Instructs an MWModulatedSource to reset all connections to their disconnected state.
Enable_Inst_Cond	Enables the instrument condition. The MWModulatedSource asserts the instrument condition to the pattern generator at the completion of a capture.
Disable_Inst_Cond	Disables an MWModulatedSource from asserting the instrument condition.

Notes on MWModulatedSource Microcodes

Note the following when using microcodes with an MWModulatedSource:

- IG-XL has rules regarding the necessary spacing in vectors between microcodes on a specific channel. If you place microcodes too close together, IG-XL will return an error. General guidelines for spacing microcodes include the following:
  - Resync: <1s
  - LoadSettingsConnect: 200 - 400s (typical)
  - Disconnect: <200s (typical)
  - Start: <10s
- An MWModulatedSource does not use PSets. It provides the equivalent capability using Signals programming in VBT.
- When using an UltraWave logical instrument (such as an MWModulatedSource), any vector that contains a POP microcode must be fully defined across all defined logical instrument instances.
  - IG-XL returns an error if you do not follow this guideline.

– Reapplying the same Signal to a given logical instrument instance results in the hardware NOPing the application. (This prevents glitches.)

For more information and an example, see Using UltraWave 12G Instruments with POP.

Use of Resync

The Resync microcode in the Microwave subsystem performs additional tasks when compared to the Resync used with the BBAC and VHFAC instruments. The primary function of Resync for BBAC and VHFAC is to align the analog clock for the analog instrument with the digital clock supplying the timing reference for the pattern.

For the Microwave instruments, the Resync microcode also:

- Forces alignment of the internal AWG clocks which drive I and Q modulation AWGs
- Synchronizes the start of waveforms

The instantiation time associated with Resync is 1 s, which you must manage. Therefore, you must place a wait of at least 1 s between the Resync and following microcode.

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UltraWave 12G MWModulatedSource Programming : MWModulatedSource Pattern Control : MWModulatedSource ASCII Pattern Syntax

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MWModulatedSource ASCII Pattern Syntax

MWModulatedSource pattern microcodes are instructions that you can place on an individual vector of a pattern. For that vector, a microcode instructs the MWModulatedSource to perform a particular action, such as starting a new signal.

You can only use MWModulatedSource microcodes with pins that the instruments statement has specified as MWModulatedSource. For example:

```
instruments = {
    DUT_AIN:MWModulatedSource;
}
```

For more information, see Instruments Statement in the Pattern Language section.

If a vector contains any MWModulatedSource related microcode other than nop, that vector is placed in SRM.

The format for specifying an MWModulatedSource microcode on a particular vector is:

(pin-list : MWModulatedSource = microcode)

The following code shows an example of ASCII pattern microcode syntax for an MWModulatedSource. The figure shows this pattern in PatternTool.

```
instruments = {
    Source:MWModulatedSource 1;
}

vm_vector
vm_MWModSourceOnly_A ( $tset      , Dig1)
{
start_label MWModSourceOnly:
```

```
                > BasicSource  0    ;
(Source:MWModulatedSource = Resync)
                > BasicSource  0    ;
repeat 10           > BasicSource  0    ; //1us wait time
(Source:MWModulatedSource = LoadSettingsConnect GSM)
                > BasicSource  0    ;
repeat 10000        > BasicSource  0    ; //1ms wait time
(Source:MWModulatedSource = Start GSM)
                > BasicSource  0    ;
repeat 100          > BasicSource  0    ; //10us wait time
(Source:MWModulatedSource = LoadSettingsConnect WCDMA)
                > BasicSource  0    ;
repeat 10000        > BasicSource  0    ; //1ms wait time
(Source:MWModulatedSource = Start WCDMA)
                > BasicSource  0    ;
repeat 100          > BasicSource  0    ; //10us wait time
(Source:MWModulatedSource = Stop)
                > BasicSource  0    ;
repeat 100          > BasicSource  0    ; //10us wait time
(Source:MWModulatedSource = Disconnect)
                > BasicSource  0    ;
repeat 2000         > BasicSource  0    ; //200us wait time
                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
```

```
                > BasicSource  0    ;
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                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
                > BasicSource  0    ;
halt            > BasicSource  0    ;
}
```

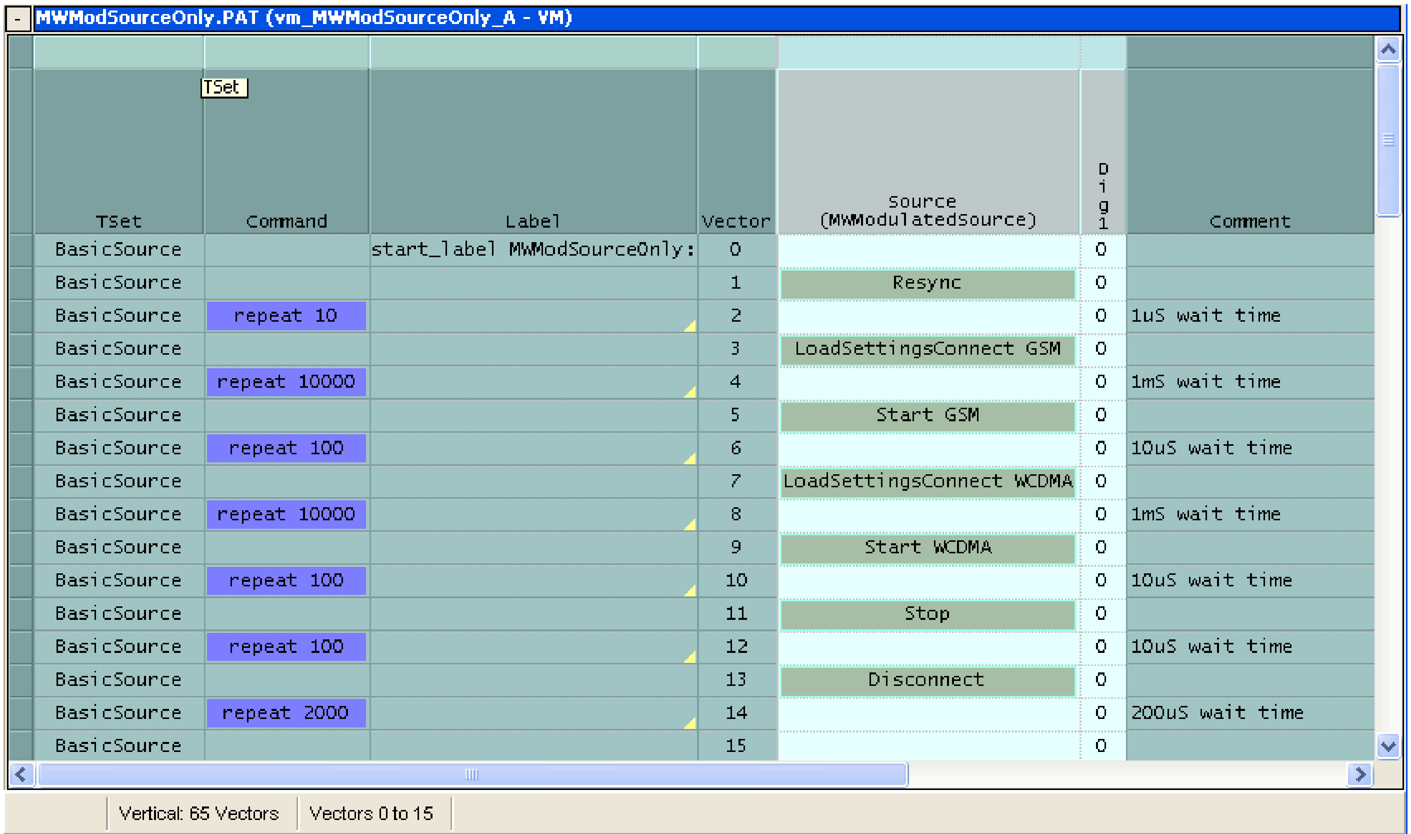
MWModSourceOnly.PAT (vm_MWModSourceOnly_A - VM)

TSet	Command	Label	Vector	Source (MWModulatedSource)	Dig1	Comment
BasicSource		start_label MWModSourceOnly:	0		0	
BasicSource			1	Resync	0	
BasicSource	repeat 10		2		0	1uS wait time
BasicSource			3	LoadSettingsConnect GSM	0	
BasicSource	repeat 10000		4		0	1mS wait time
BasicSource			5	Start GSM	0	
BasicSource	repeat 100		6		0	10uS wait time
BasicSource			7	LoadSettingsConnect WCDMA	0	
BasicSource	repeat 10000		8		0	1mS wait time
BasicSource			9	Start WCDMA	0	
BasicSource	repeat 100		10		0	10uS wait time
BasicSource			11	Stop	0	
BasicSource	repeat 100		12		0	10uS wait time
BasicSource			13	Disconnect	0	
BasicSource	repeat 2000		14		0	200uS wait time
BasicSource			15		0	

Vertical: 65 Vectors | Vectors 0 to 15

MWModulatedSource Pattern Microcodes Example

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UltraWave 12G MWModulatedSource Theory of Operation

UltraWave 12G MWModulatedSource Theory of Operation

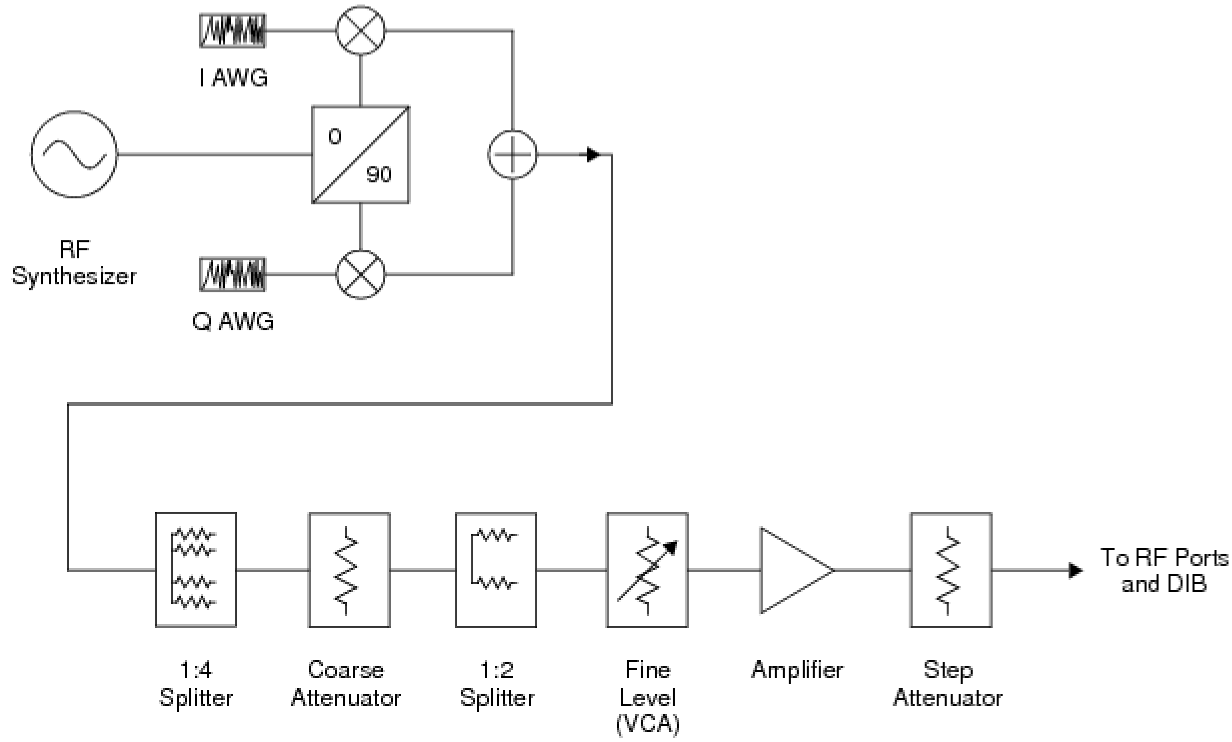
The MWModulatedSource instrument produces a programmable modulated RF signal by modulating the output of an RF frequency synthesizer. The MWModulatedSource has built-in I and Q AWGs to support both analog and digital modulation schemes.

The following topics are available:

For information on MWModulatedSource instrument performance specifications and standards, see UltraWave 12G Specifications.

Basic Operation

The figure shows a basic functional model of the MWModulatedSource.



UltraWave 12G MWModulatedSource Block Diagram

As shown, the modulated RF signal is produced by mixing an LO signal from an RF frequency synthesizer with I and Q signals from two AWGs.

Each modulated source is split for up to four Measure boards. Like the other sources, the source passes through coarse attenuation before being split for two front ends. Each source path has fine level control, amplifications, and a step attenuator.

☑ **Note:**

The signal path of a MWModulatedSource has more than one separate attenuator and amplifier. All level control elements are combined in the blocks in the diagram.

The splitters in the signal path are part of the fanout architecture of the UltraWave 12G. They allow multiple MWModulatedSources to connect to multiple RF ports and device pins, without splitters on the DIB. However, a MWModulatedSource shares a synthesizer with at least one other MWSource (through a 1:2 splitter) or more than one other MWSource on any other Measure boards (through a 1:4 splitter).

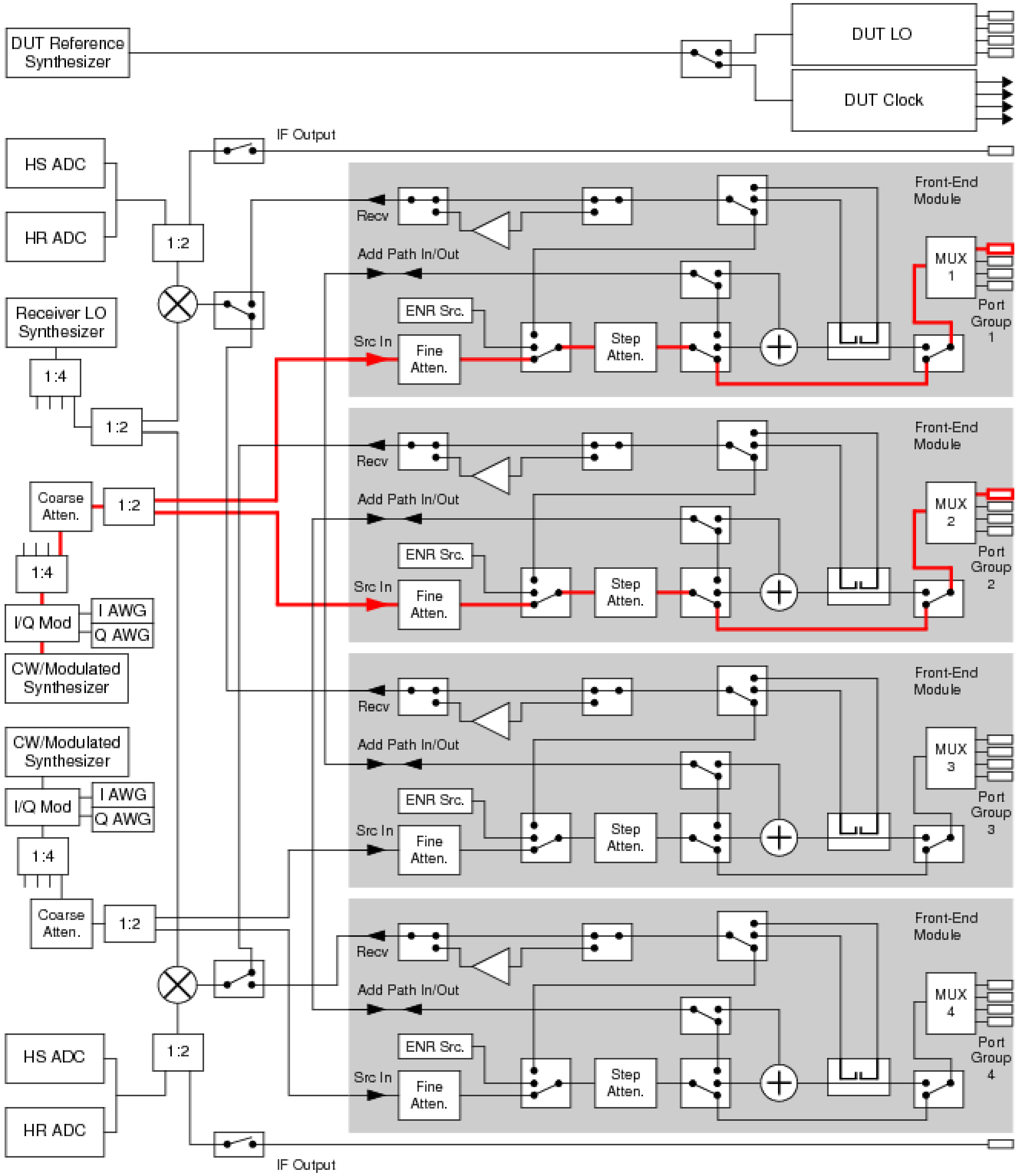
IQ Signal Input and Conversion

The MWModulatedSource uses input signals from I and Q AWGs on a Synthesizer board. I and Q signal data is loaded using the Signals interface.

Baseband signals specified as IF data are converted to IQ data by the instrument driver.

MWModulatedSource Hardware Functions and Signal Flow

The figure shows the configuration of the UltraWave 12G subsystem and signal flow for the operation and connection of two MWModulatedSources. They share an RF synthesizer. Heavy lines represent the signal flow.



MWModulatedSource Hardware Functions and Signal Path

See UltraWave 12G Hardware Functions for more information about the functions in the diagram.

For DIB connections (MW ports, Pogo pins) and signal names for the MW subsystem and instruments, see User View of UltraWave 12G SMPM Connector Blocks, User View of UltraWave 12G DIB Pad Patterns, and UltraWave 12G Signal Descriptions.

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